



Review

Insight into the influence of microbial aggregate types on nitrogen removal performance and microbial community in the anammox process - A review and meta-analysis



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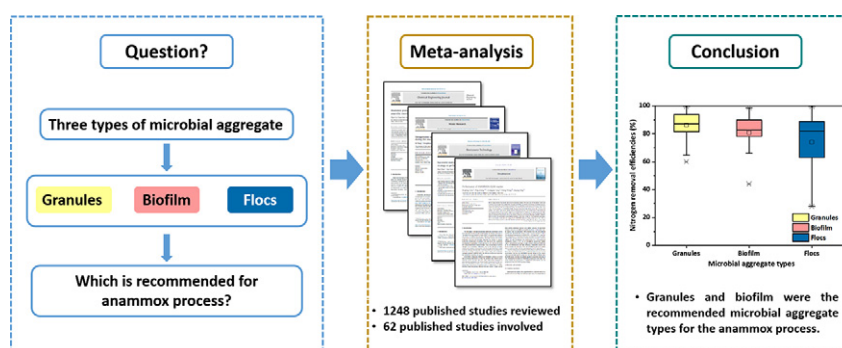
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HIGHLIGHTS

- Meta-analysis indicated that the granular sludge appeared to be competent in achieving the highest average nitrogen removal efficiency of 81.1%, followed by biofilm of 80.8% and flocs of 74.1%.
- The relative abundance and diversity of the microbial community in granules and biofilm were higher than flocs.
- Granules and biofilm were the recommended microbial aggregate types for the anammox process.

GRAPHICAL ABSTRACT



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ABSTRACT

The anaerobic ammonium oxidation (anammox) process has been paid close attention in the wastewater treatment field because of its energy-saving advantages. Different microbial aggregates have been used in the anammox process, and there is an urgent need to evaluate the comparative efficiencies of the widely used types of microbial aggregates with respect to their nitrogen removal performance as well as microbial community. To address this, 1724 published papers concentrating on three types of microbial aggregates, namely granules, biofilm, and flocs were compiled. A quantitative meta-analysis was carried out to compare the standard error of nitrogen removal efficiencies among these three microbial aggregates. The data sources of this meta-analysis comprised articles on granules (42%), followed by those on biofilm (33%) and flocs (25%). The granular sludge appeared to be competent in achieving the highest average nitrogen removal efficiencies of 81.1%, followed by biofilm (80.8%). Flocs provided comparatively poor removal of nitrogen pollutants with the lowest removal efficiency of 74.1%. Biofilm had the highest abundance of functional microbial communities with 43.4% on *Candidatus Kuenenia* and 11.2% on *Candidatus Brocadia*, which were detected in the anammox system as common genera. This meta-analysis suggested that the microbial aggregate types of granules and biofilm had a relatively low heterogeneity and high total nitrogen removal efficiencies for the anammox process and were the recommended microbial aggregates for anammox bacteria cultivation and operation of the anammox process.

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Nomenclature	
AHLs	N-acylated-L-homoserine lactones
anammox	anaerobic ammonium oxidation
c-di-GMP	bis(3'-5')-cyclic dimeric guanosine monophosphate
EPS	extracellular polymeric substances
HRT	hydraulic retention time
NAD	nicotinamide adenine dinucleotide
NLR	nitrogen loading rate
NRE	nitrogen removal efficiency
NREs	nitrogen removal efficiencies
NRR	nitrogen removal rate
PN	protein
PN/A	partial nitrification-anammox
PS	polysaccharides
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
QS	quorum sensing
SAA	specific anammox activity
SBR	sequencing batch reactor
SBBR	sequencing biofilm batch reactor
SE	standard error
SMD	standard mean difference
SRT	solids retention time
TN	total nitrogen
UASB	upflow anaerobic sludge blanket

1. Introduction

Anaerobic ammonium oxidation (anammox) is a noted biological technology process for nitrogen removal and has been accepted as a promising and sustainable technology in wastewater treatment (Guo et al., 2017b; Zhang et al., 2017). It has received wide attention because of its environment-friendly and energy-saving features.

In anammox reactors, the microorganisms can form three types of microbial aggregates, granules (Sobotka et al., 2017; Xiong et al., 2013), biofilm (Y. Sun et al., 2018; Suto et al., 2017), and flocs (Zheng et al., 2016). Bioaggregation can facilitate solid-liquid separation and increase solid retention time (Sheng et al., 2010). However, there are also problems with each type of bioaggregation. One of the most frequent problems with sludge flocs is the low settling rate and irreversible

breakage (Wang et al., 2017). The small flocs and cell fractions cannot settle well in the downstream unit and enter the effluent and cause an increase in turbidity (Zhang et al., 2007). As for biofilms, typical problems are mass transport limitation and that only the bacteria in the outer layers of the biofilm contribute to overall substrate removal (Henze et al., 2008). In addition, the cost of biofilm carriers is also a limitation for its application in the large-scale wastewater treatment plant (Zhao et al., 2019). Compared to flocs and biofilms, granules have a dense structure and need not attach to the surface of a media. However, the disadvantage of anammox granules is a long start-up period owing to the low cellular yield and growth rates of anammox bacteria (Kartal et al., 2012; Tay et al., 2006). A previous study revealed that it took 98 days to start up the anammox granule reactor by seeding denitrifying sludge (Li et al., 2012). Similarly, Chen et al. (2016) started an anammox UASB reactor successfully on Day 91, while the start-up time was shortened to 40 days by seeding the denitrifying-anammox sludge.

Nitrogen removal performance of the anammox process with the different microbial aggregates showed high variations. The NREs of granular sludge processes varied from 60% (Cho et al., 2010) to 99.7% (Emilia Rios-Del Toro et al., 2017). Moreover, the NREs of biofilm process ranged from 44% (Tsushima et al., 2007) to 98.7% (Jin et al., 2008), whereas for flocs process, the NREs changed from 28% (Kieling et al., 2007) to 99.7% (Ismail et al., 2019). These different findings usually attributed to differences in temperature.

Such inconsistencies highlight our incomplete understanding of the effect of different microbial aggregate types on NREs of the anammox process. It is necessary to evaluate and form a conclusion on the NREs of different microbial aggregate types in the anammox process using quantitative assessment. This will help determine which type of microbial aggregate is the best for the improvement of process efficiency.

The nitrogen removal performance is related to the microbial community in the anammox reactors. Recently, high-throughput sequencing, one of the most advanced techniques, has been widely used to study the diversity and community of microorganisms in anammox reactors at a deeper level without culturing (Polka et al., 2015; Yang et al., 2018; Zheng et al., 2016). Nineteen species associated with six genera in the taxonomic group of anammox microorganisms have been recognized (Schmid et al., 2005). Six genera which are characterized for anammox biomass have been presented as *Candidatus Kuenenia*, *Candidatus Brocadia*, *Candidatus Jettenia*, *Candidatus Scalindua*, *Candidatus Anammoximicrobium* and *Candidatus Anammoxoglobus* (Kuenen, 2008; Li et al., 2018).

The abundances and species of microbial communities are mostly influenced by the microbial aggregates and operational parameters, such as temperature, HRT, NLR, and SRT (Helmut et al., 2011; Khammar et al., 2005; Tomaszewski et al., 2017; Vlaeminck et al.,

2010). *Candidatus Brocadia* was found in an anammox upflow granular sludge bed reactor (Sobotka et al., 2017). In a pilot-scale PN/A reactor, *Candidatus Brocadia* and *Candidatus Jettenia* were observed in granules (Wang et al., 2017). *Candidatus Kuenenia* was found in both SBBR and non-woven fabric membrane reactors, achieving the relative abundance of total bacteria of 12.38% and 8.1%, respectively, of total bacteria (Miao et al., 2018; Ren et al., 2018). Moreover, *Candidatus Brocadia* together with *Candidatus Jettenia* were explored in a biofilm reactor Ren et al. (2018), and the proportions were 11.2% and 2.7%, respectively. For the reactor incubated with flocs, it was found that the *Candidatus Kuenenia* had relative abundance of 15.7% (Wu et al., 2018). Kartal et al. (2007) found that in the presence of ammonium, nitrate and nitrite, *Candidatus Anammoxoglobus propionicus* was gathered in an SBR and had the ability to out-compete other bacteria, including other genera of anammox and heterotrophic denitrifiers, for propionate oxidation.

In addition, operational parameters could also influence the abundance and species of microbial communities. Temperature can affect the abundance and structure of anammox species. As reported before, *Candidatus Brocadia* preferred temperatures of >30 °C, while *Candidatus Kuenenia* became competitive at 13 °C, reaching the proportion of 55.18% of the microbial community, in an UASB reactor (He et al., 2018). Moreover, it has been observed that *Candidatus Brocadia sinica* was predominant with a relative abundance of 0.5% in the bioreactors irrespective of the kinds of seeding sludge inoculated at the NLR of 1 kg N m⁻³ d⁻¹ (Cho et al., 2018). The relative abundance of *Candidatus Brocadia* was 2.37% in an UASB with NRR of 2.39 kgN m⁻³ d⁻¹ at ambient temperature (Cao et al., 2016). Interestingly, the granular sizes also affected anammox bacterial diversity, and *Candidatus Brocadia* was the unique anammox genus found in granules below 0.5 mm, and *Candidatus Kuenenia* and *Candidatus Brocadia* were detected in granular sludge over 0.9 mm in size (Zhu et al., 2018).

To better understand the comparative efficiencies of these processes, a meta-analysis was conducted to quantitatively investigate the effect of different microbial aggregates on the NRE of the anammox process. The correlations between the relative abundances of anammox microorganisms and operational parameters were also investigated to gain a comprehensive understanding of the anammox process.

2. Materials and methods

2.1. Data sources

After several trials on different search terms, the search work was conducted using the online scholarly databases (PubMed and Web of Science). No time restriction was set for the search. Moreover, no language restriction was applied in the search strategy. The key words of this research were as follows: TOPIC: (anammox or anaerobic ammonium oxidation) AND TOPIC: (nitrogen removal or nitrogen removal rate) AND TOPIC: (granular or granules or biofilm or flocs or suspended). The search was conducted on July 25, 2019. After this initial selection, the titles and abstracts were carefully reviewed to determine whether the studies met the following inclusion criteria:

- (1) Studies should be written in English and published in journals in Science Citation Index;
- (2) The process should not combine anammox with other nitrogen removal pathways, e.g., partial nitrification, denitrification or denitrifying anaerobic methane oxidation;
- (3) Studies should not include the effect of various toxic substrates (such as heavy metals and thiocyanate) or other operational parameters (e.g., pH, temperature) on the anammox process;
- (4) The nitrogen concentrations should be clearly presented.

The selection process is summarized in a flow chart (Fig. 1).

2.2. Data extraction and analysis

The data for meta-analysis were obtained from each selected study, including reactor types, microbial aggregate types, NREs, HRT and temperature. Information on the subsequent outcomes was acquired from individual trials as well as the online supplementary material.

When the NRE was not directly provided in the study, NRE could be calculated as follows (Eq. (1)).

$$NRE(\%) = \frac{C_{influent} - C_{effluent}}{C_{influent}} \times 100 \quad (1)$$

Where $C_{influent}$ and $C_{effluent}$ were the concentrations of the influent and effluent TN, respectively.

To compare NRE among the microbial aggregate types, the removal data was converted to an SE for each experiment independently. SE was the difference in NRE in each experiment (X) compared with the mean NRE ($\bar{x}$) in different microbial aggregate types. The SE was calculated as follows (Eq. (2)):

$$SE = \sqrt{1/X + 1/|n - X|} \quad (2)$$

The SE for each experiment (for NRE) was calculated and analyzed for significant differences in the mean SEs (for NREs) using randomizations without replacement.

2.3. Statistical analysis

Meta-analysis, a method of quantitative analysis, is the scientific integration of outcomes in relative research. In meta-analysis, results are extracted from each study in terms of effect sizes and further applied to combine the results from multiple published studies using the same scale, utilizing suitable metrics; SMD was conducted in this study (Nakagawa and Santos, 2012). The SMD, which was defined as the difference in each NRE and the total means of NRE divided by their common standard deviation was calculated by pooling the standardized mean improvement of each study using Review Manager 5.3. The effect sizes are added into the statistical model to assess entire effects and heterogeneity in outcomes (Hedges and Olkin, 2014). To accommodate the anticipated heterogeneity across studies, the random-effects model was applied to support a summary effect size which represented the mean of the distribution of effects from the included studies (Riley et al., 2011).

During the meta-analysis, it was also important to investigate the central tendency (the mean) associated with the confidence limits, and the heterogeneity in effect across studies. These procedures were conducted to maintain the accuracy of meta-analysis. In the meta-analysis of each outcome, a pre-planned sensitivity analysis was conducted to restrict the trials that assess the influence of microbial aggregate types on NREs. As the between-study heterogeneity is often shown in meta-analysis, it is necessary to perform the tests of publication bias in such situations. Statistical heterogeneity was evaluated by the Cochran Q test as well as the I^2 statistic (Higgins et al., 2003). The Higgins I^2 statistics was utilized to determine the degree of between-study heterogeneity (I^2 , low: < 25%, moderate: 25%–50%, high: > 50% of heterogeneity) (Higgins and Thompson, 2002). The statistical Q tests and I^2 tests were performed in Review Manager 5.3 (The Cochrane Collaboration, 2014).

The Pearson correlation matrix was conducted to evaluate the correlation of the relative abundance of anammox bacteria, NRE, NLR, temperature and HRT. SPSS Statistics 20 was used for the analysis.

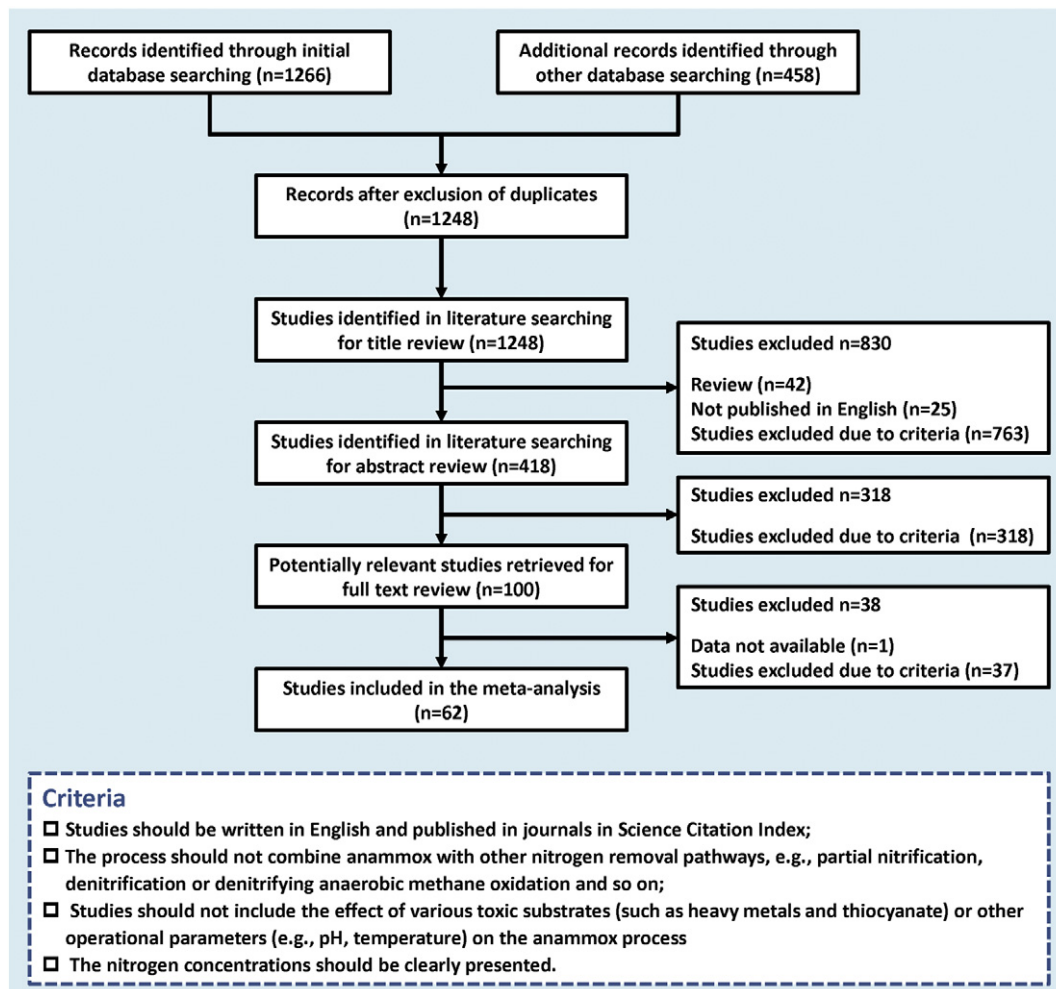


Fig. 1. The flow diagram of this research, selection process and criteria.

3. Results and discussion

3.1. Data collection and compilation

The literature research yielded 1724 studies in total. Exclusion of repetitive studies resulted in an initial list of 1248 studies. The principle defined by the PRISMA method was followed to further choose the satisfactory results. After exclusion the review articles, all the abstracts or full-texts of 1206 studies were read to decide whether the research was related to the nitrogen removal process of the anammox bacteria or not.

According to the criteria mentioned in 2.1, a total of 62 articles (shown in supplementary materials, Table S1) were identified and screened. Of these 62 articles, 42% were on granule type of microbial aggregates, and 33% on were on biofilms and 25% on flocs.

3.2. Meta-analysis

Meta-analysis, a quantitative analysis, is the scientific integration of outcomes in relative research. Forest plot, shown in supplementary materials (Fig. S1), was used to graphically describe the effect sizes as well as their confidence intervals (Gurevitch et al., 2018).

After the outliers were excluded, the pooled SMD for the granules, biofilm, and flocs were 0.1 (95% CI: -0.18 – 0.73 ; $P < 0.00001$), -0.00 (95% CI: -0.60 – 0.59 ; $P < 0.00001$) and -0.01 (95% CI: -0.53 – 0.52 ; $P < 0.00001$), respectively (Fig. S1). The granules, biofilm, and flocs showed considerable heterogeneity of 31%, 46%, and 50%, respectively,

suggesting low-to-moderate levels of heterogeneity. Relatively low-to-moderate heterogeneity in our meta-analysis indicated that the target parameters based on the NREs were probably highly comparable with each other, suggesting the rationality for further analysis.

3.3. NREs of three types of microbial aggregate

A boxplot is a visible graphical tool, which is widely applied in exploratory data analysis (Abuzaid et al., 2012). Boxplots were used to display the NREs of three microbial aggregate types (Fig. 2). The average NREs of granules, biofilm, and flocs were 81.1%, 80.8%, and 74.1% respectively. It was observed that the microbial aggregate of granules or biofilm achieved higher NREs than of flocs. The bottom and top values in the boxes represents the quartiles (25% and 75% percentile) and line inside the box represents the median. The interquartile range with upper (75%) and lower (25%) quartiles of three microbial aggregate types, granules, biofilm and flocs, were 94.2% and 89.8%, 89.0% and 81.1%, and 77.7% and 57.5%, respectively. It suggested that granules had stable data, associated with a box with a slight altitude of NREs, with the proportion of 4.4%, while flocs had a wide range of 20.2%. In general, the granules and biofilm types of microbial aggregates had higher and more stable nitrogen removal performance than flocs.

The better performance of the granules and biofilms processes can be attributed to the longer biomass retention time which is also considered in terms of sludge age and average cell retention time (Chen et al., 2016). For anammox bacteria, previous studies have demonstrated that the minimum doubling time was approximately 11 d and the specific

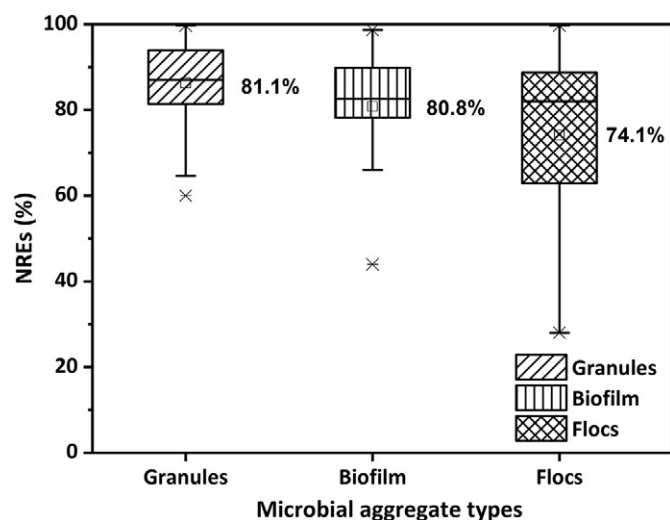


Fig. 2. NREs of three types of microbial aggregate (Granules, $n = 37$; biofilm, $n = 26$; flocs, $n = 21$). Boxplots display interquartile range, min and max (whiskers), median (horizontal line) and average ($\square$) of NREs in the final dataset.

growth rate was slow, approximately to $0.0027 \pm 0.005 \text{ h}^{-1}$ (Van de Graaf et al., 1996). Therefore, the development of the anammox community preferred longer SRT and SRT was also an important parameter. When the SRT was higher than the doubling time of anammox bacteria, a good performance could be achieved (Chen et al., 2016). Therefore, granule and biofilm type of microbial aggregates could provide longer retention time as long as SRT for anammox bacteria.

It has been demonstrated that granules had a more regular shape and a higher biomass density than that of the flocs (Yu and Joo-Hwa, 2004). In addition, previous studies had reported that the equivalent diameters were around 0.5–4 mm and densities were in the range of $1.04\text{--}1.05 \text{ kg m}^{-3}$, comparable with that of flocs that were $1.02\text{--}1.06 \text{ kg m}^{-3}$ (Dammel and Schroeder, 1991; de Kreuk and Van Loosdrecht, 2004; Gobi et al., 2011; Shi et al., 2009). Granules with geometry and free movement characteristics limited resistances of the external boundary layer, improving mass transfer of substrate towards the microorganisms (Baeten et al., 2019). However, the oversized granules might decay and leak out inert cellular products because of substrate limitations (Ni et al., 2009). The model experiment by Ni et al. (2009) showed that the effluent nitrogen concentrations decreased and NREs increased with the granule diameter increasing from 0.25 mm to 2.75 mm. Model simulations for the anammox granules also showed that the optimum granule diameter for maximum nitrogen removal in such cases should be between 1.0 mm and 1.3 mm and that the optimum NLR should be $0.8 \text{ kg N m}^{-3} \text{ d}^{-1}$ (Ni et al., 2009). In general, the constitution of anammox granules could accelerate the mass transfer and achieve longer SRT and overcome the disadvantages in anammox bacteria, e.g., longer doubling time as well as a lower growth yield, which contributed to higher NREs.

Biofilm are a kind of accumulation microorganisms that often attach to and cover the solid surface (Characklis and Marshall, 1990). They exhibit a longer SRT because of the improved physical retention and potentially higher biodiversity than that of flocs which can present an option to improve micropollutant removal performance (Torres et al., 2016). The biofilms were also particularly efficient at protecting microorganisms which grew slowly from being washed out (Woolard, 1997).

QS, which is a bacteria communication mechanism mediated by autoinducers, is also known to regulate nitrogen removal. QS can influence gene expression and regulate the metabolism of a bacterial community (Tang et al., 2018; Zhu et al., 2002). Recently, AHLs and *c-di-GMP* as common kinds of autoinducers, were detected in the anammox system (Guo et al., 2017a; Y. Sun et al., 2018; Sun S. et al., 2018; Tang et al.,

2018). The secretion of the autoinducers was correlated to bacteria activities and could mediate EPS synthesis, and hence influenced NRE.

QS is a density dependent mechanism for bacterial communication. The cell densities of granules and biofilm were much higher than flocs (Figdore et al., 2018; Ren et al., 2018), and the bacteria communication in granules and biofilm could be strengthened (Kartal et al., 2012; Strous et al., 1999) and higher AHLs concentrations could be detected. It had been reported that more AHLs existing in granules than in flocculent seeding sludge (Chen et al., 2019; Lv et al., 2014). There were also more types and concentrations of AHLs in mature biofilm than in activated sludge (Hu et al., 2016; Li et al., 2019).

QS could regulate the gene expression. It was demonstrated that AHLs could influence the expression of *hzsA*, a genetic marker to evaluate the activity of anammox bacteria (Ali et al., 2014). The expressional level of *hzsA* increased by 24% with exogenous addition of $2 \mu\text{M}$ *N*-3-oxo-hexanoyl-L-homoserine to anammox system (Tang et al., 2019). The change of expression level of *hzsA* increased SAA by 8% (Tang et al., 2018).

QS could also regulate metabolic pathway. It had been confirmed that LysoPC (20:0) participated in metabolism of glycerophospholipid (lipid metabolism) which had positive effect on the biomass yield rate of anammox bacteria (Tang et al., 2018). As reported before, NAD was of great importance in anammox activity because it frequently took part in redox reactions and mainly regulated the electron transport carriers, electron-transport chain from the citrate cycle to the respiratory (Tang et al., 2018; Watanabe et al., 2009). AHLs, e.g., *N*-(3-oxohexanoyl)-DL-homoserine lactone, *N*-hexanoyl-DL-homoserine lactone and *N*-octanoyl-DL-homoserine lactone, were found to increase the biosynthesis of NAD and LysoPC (20:0) (Tang et al., 2018). It further increased SAA by 8%, 11%, and 7%, respectively (Tang et al., 2018). The increase of SAA could cause the enhancement of nitrogen removal (Tang et al., 2014).

Overall, higher QS activity in granules and biofilm than in flocs contributed to the higher anammox activity, and the improvement of the anammox bacteria activity led to the better nitrogen removal performance of the granules as well as biofilm processes.

In addition, QS was related to EPS production. AHLs could regulate the production of amino acids, e.g., Ala, Val, Glu, Asp and Leu, to improve the PN production in EPS in anammox cells (Tang et al., 2018). AHLs could mediate UDP-GlcNAc pathway to increase PS production in EPS. It had been confirmed that exogenous AHLs could promote EPS production in both mature biofilms (Hu et al., 2016) and granular sludge (Ma et al., 2018). Besides, *c-di-GMP*, a second messenger, had been reported to promote the EPS production of *Candidatus Jettenia caen* which was a kind of anammox species (Guo et al., 2017b).

A higher EPS content in microbial aggregates resulted in better microbial aggregates stability (Mikkelsen and Nielsen, 2001), which would contribute to the stable nitrogen performance. EPS were beneficial for maintenance of granules stability (Long et al., 2014; McSwain et al., 2005). The EPS could protect the granules against shear forces and gas bubbles effectively which maintained the stability of the granules (Pol et al., 2004). EPS could serve as source anchorage for microorganisms and help maintain the stability of biofilm (Sgier et al., 2018). Higher concentrations of EPS, especially PN, had the ability to support a rigid net for functional bacteria, and then the excellent pollutant removal property was formed (Zhang et al., 2019). Generally, microbial aggregates of granules and biofilms with higher EPS content than that in flocs resulted in higher and more stable NREs.

Interestingly, previous researches had found that EPS took part in the electron transport processes (Xiao and Zhao, 2017). The electron mediators in EPS, e.g., PN, PS, humic substances, glycoproteins, and in some cases, extracellular DNA, might be beneficial for cell growth and microbial metabolism via the extracellular electron transfer function (Xiao and Zhao, 2017). The higher EPS concentrations in granules and biofilm might stimulate the extracellular electron transfer process so that the NREs could be enhanced.

Overall, QS activities are higher in granules and biofilm systems, and bacteria communication could be strengthened among them. QS

Table 1
Microbial community species and relative abundance in published papers in meta-analysis.

No.	Source	Microbial aggregate types	Microbial species in genus level	Relative abundance
1	(Cao et al., 2016)	Granules	<i>Candidatus Brocadia</i>	2.37%
			<i>Candidatus Kuenenia</i>	0.03%
2	(Chen et al., 2017)	Granules	<i>Candidatus Jettenia</i>	4.92%
3	(Ni et al., 2019)	Granules	<i>Candidatus Brocadia</i>	17.1%
			<i>Candidatus Kuenenia</i>	12.6%
4	(Ni et al., 2019)	Granules	<i>Candidatus Brocadia</i>	7.1%
			<i>Candidatus Kuenenia</i>	9%
5	(S. Sun et al., 2018)	Biofilm	<i>Candidatus Kuenenia</i>	39.91%
6	(S. Sun et al., 2018)	Biofilm	<i>Candidatus Kuenenia</i>	37.99%
7	(Wu et al., 2018)	Biofilm	<i>Candidatus Kuenenia</i>	43.4%
			<i>Candidatus Brocadia</i>	11.2%
8	(Zhang et al., 2018)	Biofilm	<i>Candidatus Kuenenia</i>	12.2%
9	(Zhang et al., 2018)	Biofilm	<i>Candidatus Kuenenia</i>	20%
11	(Wu et al., 2018)	Flocs	<i>Candidatus Kuenenia</i>	15.7%

regulates the bacteria activity and mediates EPS production, so that the higher and more stable NREs are obtained.

3.4. Microbial community analysis

Among the studies in this meta-analysis, 11 studies have reported the microbial community species abundance based on high-throughput sequencing technology. The microbial community species as well as the relative abundance mentioned in the published papers were shown in Table 1. The prevalent genera in anammox bacteria were mainly *Candidatus Brocadia*, *Candidatus Kuenenia* and *Candidatus Jettenia*. The granules and biofilm types of microbial aggregates had higher relative abundances of microbial communities than flocs. The highest relative abundance of 44% of *Candidatus Brocadia* was achieved in the granules type of microbial aggregates. The most relatively abundant anammox species were observed in the biofilm type of microbial aggregates, with 43.4% of *Candidatus Kuenenia* and 11.2% of *Candidatus Brocadia*. The relative abundance of anammox species in flocs was at most, 15.7%.

The bacteria of anammox have consistently belonged to the family, *Brocadaceae*. At genus level, the species *Candidatus Brocadia*, *Candidatus Kuenenia*, and *Candidatus Jettenia* could be commonly found in the anammox systems, which have been reported before (Hu et al., 2010). It had been suggested that *Candidatus Kuenenia* might be a k-strategist (e.g., high substrate affinity associated with relatively low growth rate) and *Candidatus Brocadia* might be an r-strategist (e.g., low substrate affinity combined with relatively high growth rate) (Isanta et al., 2015; Van Der Star et al., 2008).

Based on Table 1, results of the Pearson correlation analysis among the relative abundance of microbial community and other operational

parameters were illustrated in Fig. 3. The relationship between NREs and the abundance of the microbial community had not significant correlation. It was observed that temperature had a negative correlation with the relative abundance of microbial community. The higher temperature did not favor the abundance of the microbial community. As reported before (Table S1), the appropriate temperature for the majority of anammox species was in the range of 28 °C to 37 °C. Interestingly, researchers also found that *Candidatus Kuenenia* was dominant at relative low temperature of 13 °C, accounting for 55.18% of the microbial community (He et al., 2018). However, there were some studies which demonstrated that the suitable temperatures for the growth of *C. Brocadia sinica* and *C. Kuenenia stuttgartiensis* were 25 °C–45 °C (Egli et al., 2001) and 25 °C–37 °C (Oshiki et al., 2011), respectively. Researchers have confirmed that higher temperature is beneficial for the anammox activity (Dosta et al., 2008; He et al., 2018). It has been reported that the anammox activity dropped by 64.8% when the temperature switched from 30 °C to 20 °C (De Cocker et al., 2018). In general, the relatively high temperature benefitted anammox activity (Zhou et al., 2014) and favored the NREs (He et al., 2018), however, it decreased the abundance of the microbial community in the anammox system. The relative abundance of the microbial community might not necessarily have a positive effect on NREs, and the microbial activity might be more important for NREs.

4. Limitations

This study has few limitations. First, although 1248 published articles were reviewed, the inclusion criteria resulted in only 62 publications. It was a relatively small sample of the search results. The search strategy should be expanded to other treatment processes of wastewater treatment technologies (e.g., nitrification, denitrification).

Second, studies in this meta-analysis were carried out a period of 20 years, a relative long period during which approaches of detection have evolved. Moreover, the data contained in this meta-analysis were extracted from various laboratories which used different sample processing methods and measurement technologies.

Thirdly, the types of reactor were varied in different laboratories. The shortage of the positive or negative controls was a major challenge in this study.

Last but not the least, because of the limitation of meta-analysis, operational parameters, e.g., HRT, temperature, NLR, and others, were not considered.

5. Conclusions

- (1) A comprehensive meta-analysis was adopted to quantitatively understand the influence of the type of microbial aggregates on nitrogen removal performance of the anammox process. The meta-analysis results showed that the average NREs of granules,

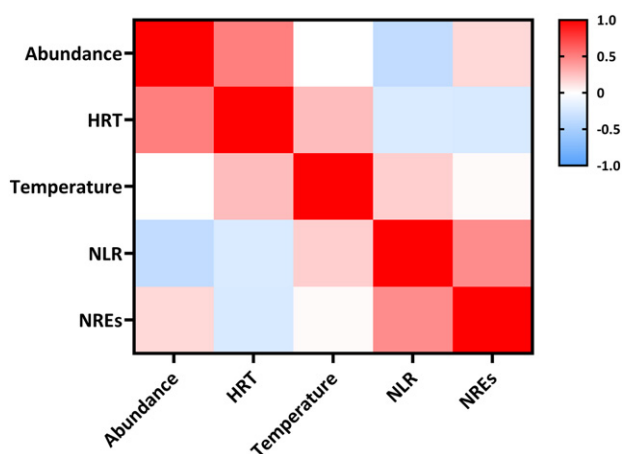


Fig. 3. Pearson correlation analysis results among the abundance of microbial community and HRT, temperature, NLR, NREs.

biofilm, and flocs processes were 81.1%, 80.8%, and 74.1% respectively. It was observed that granules as well as biofilm processes had higher and more stable NREs than flocs. The granules and biofilm processes are the recommended types of microbial aggregates for the anammox process.

- (2) The analysis of microbial community revealed that *Candidatus Brocadia*, *Candidatus Kueneia*, and *Candidatus Jettenia* were the most dominant genera during the anammox process. The relative abundance and diversity of the microbial community in granules and biofilms were higher than that in flocs. Moreover, higher temperature decreased the proportion of the microbial community in the anammox system.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2020.136571>.

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